Hamilton takes the cyclical challenges offered by parasites and makes them the basis for an altogether grander theory, his theory of why sex exists at all. But here we are concerned with his use of parasites to solve the paradox of vanishing variation in sexual selection. He believes that hereditary disease-resistance among males is the most important criterion by which females choose them. Disease is such a powerful scourge that females will benefit greatly from any ability they may have to diagnose it in potential mates. A female who behaves like a good diagnostic doctor and chooses only the healthiest male for mate will tend to gain healthy genes for her children. Now, because the definition of the 'best rabbit' is always changing, there will always be something important for females to choose between, when they look the males over. There will always be some 'good' males and some 'bad' males. They won't all become 'good' after generations of selection, because by then the parasites will have changed and so the definition of a 'good' rabbit will have changed. Genes for resisting one strain of myxoma virus will not be good at resisting the next strain of myxoma virus that mutates on to the scene. And so on, through indefinite cycles of evolving pestilence. Parasites never let up, so females cannot let up in their relentless search for healthy mates.

How will the males respond to being scrutinized by females acting as doctors? Will genes for faking good health be favoured? To begin with, maybe, but selection will then act on females to sharpen up their diagnostic skills and sort out the fakes from the really healthy. In the end, Hamilton believes, females will become such good doctors that males will be forced, if they advertise at all, to advertise honestly. If any sexual advertisement becomes exaggerated in males it will be because it is a genuine indicator of health. Males will evolve so as to make it easy for females to see that they are healthy—if they are. Genuinely healthy males will be pleased to advertise the fact. Unhealthy ones, of course, will not, but what can they do? If they don't at least *try* to display a health certificate, females will draw the worst conclusions. By the way, all this talk of doctors would be misleading if it suggested that females are interested in curing males. Their only interest is diagnosis, and it is not an altruistic interest. And I'm assuming that it is no longer necessary to apologize for metaphors like 'honesty' and 'drawing conclusions'.

To return to the point about advertising, it is as though males are forced by the females to evolve clinical thermometers permanently sticking out of their mouths, clearly displayed for females to read. What might these 'thermometers' be? Well, think of the spectacularly long tail of a male bird of paradise. We have already seen Fisher's elegant explanation for this elegant adornment. Hamilton's explanation is altogether more down-to-earth. A common symptom of disease in a bird is diarrhoea. If you have a long tail, diarrhoea is likely to mess it up. If you want to conceal the fact that you are suffering from diarrhoea, the best way to do it would be to avoid having a long tail. By the same token, if you want to advertise the fact that you are *not* suffering from diarrhoea, the best way to do so would be to have a very long tail. That way, the fact that your tail is clean will be the more conspicuous. If you don't have much of a tail at all, females cannot see whether it is clean or not, and will conclude the worst. Hamilton would not wish to commit himself to this *particular* explanation for bird of paradise tails, but it is a good example of the *kind* of explanation that he favours.

I used the simile of females acting as diagnostic doctors and males making their task easy by sporting 'thermometers' all over the place. Thinking about other diagnostic standbys of the doctor, the blood pressure meter and the stethoscope, led me to a couple of speculations about human sexual selection. I'll briefly present them, though I admit that I find them less plausible than pleasing. First, a theory about why humans have lost the penis bone. An erect human penis can be so hard and stiff that people jokingly express scepticism that there is no bone inside. As a matter of fact lots of mammals do have a stiffening bone, the baculum or os penis, to help the erection along. What's more, it is common among our relatives the primates; even our closest cousin the chimpanzee has one, although admittedly a very tiny one which may be on its evolutionary way out. There seems to have been a tendency to reduce the os penis in the primates; our species, along with a couple of monkey species, has lost it completely. So, we have